

137 million ED visits: what a national sample tells hospital managers

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21,061

sampled visits (rows) from 248
EDs, 1,031 variables

136.9M

U.S. ED visits once weighted by
PATWT, exactly the NCHS total

-9 / -8 / -7

blank / unknown / N/A codes,
hidden missing data we removed

20 / 20

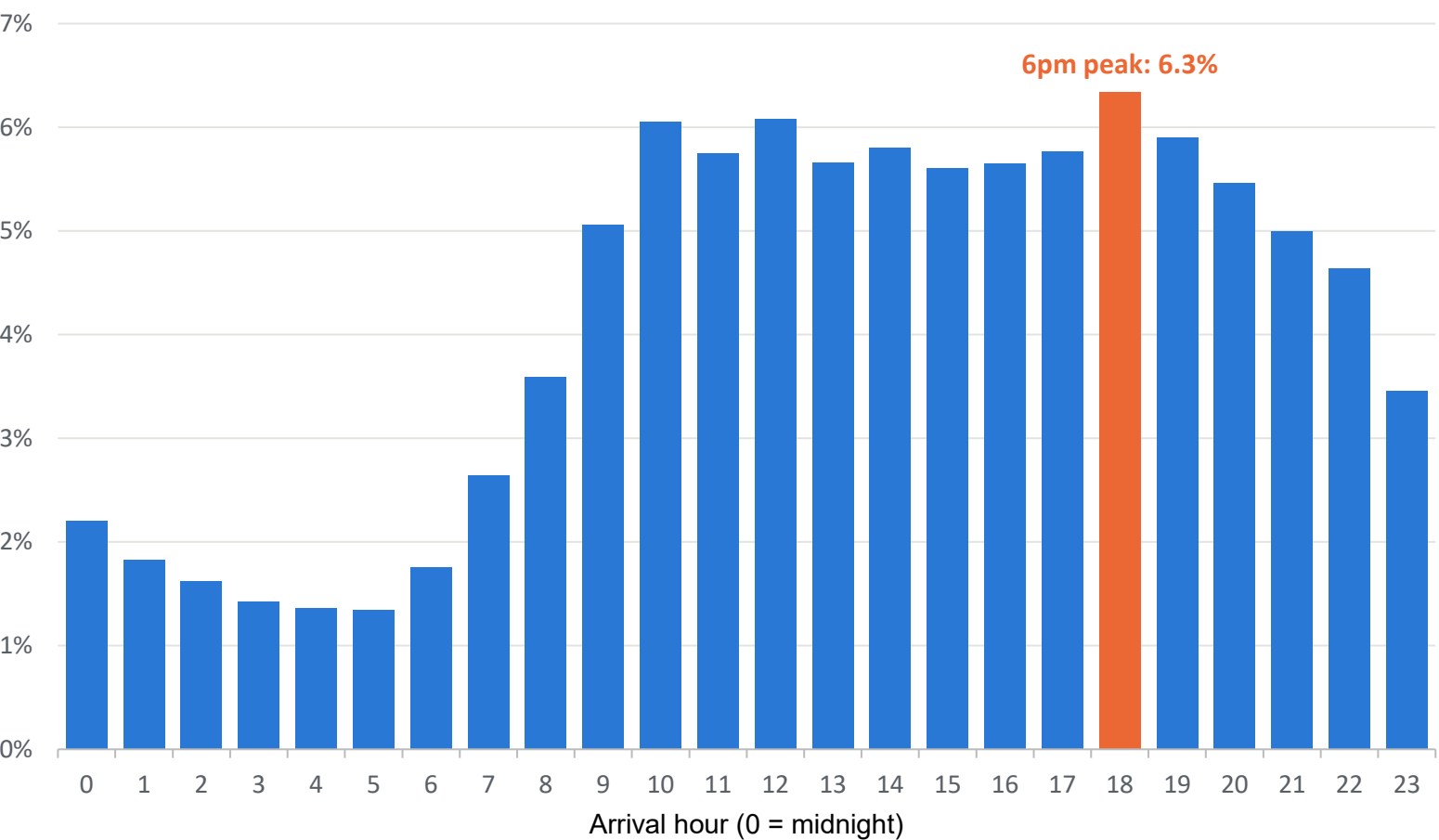
checks against published NCHS
figures pass

Approach

Load → recode missing-value codes → weight every estimate → cap extreme waits and stays at the 99.5th percentile → check each result against the codebook. One commented notebook, one cell per question, re-run cleanly end to end.

Demand is predictable: 10am-8pm plateau, Monday peak

Weighted share of daily ED arrivals by hour of arrival, 2015



4.7x

arrivals in the 6pm peak hour vs the 5am trough

+25%

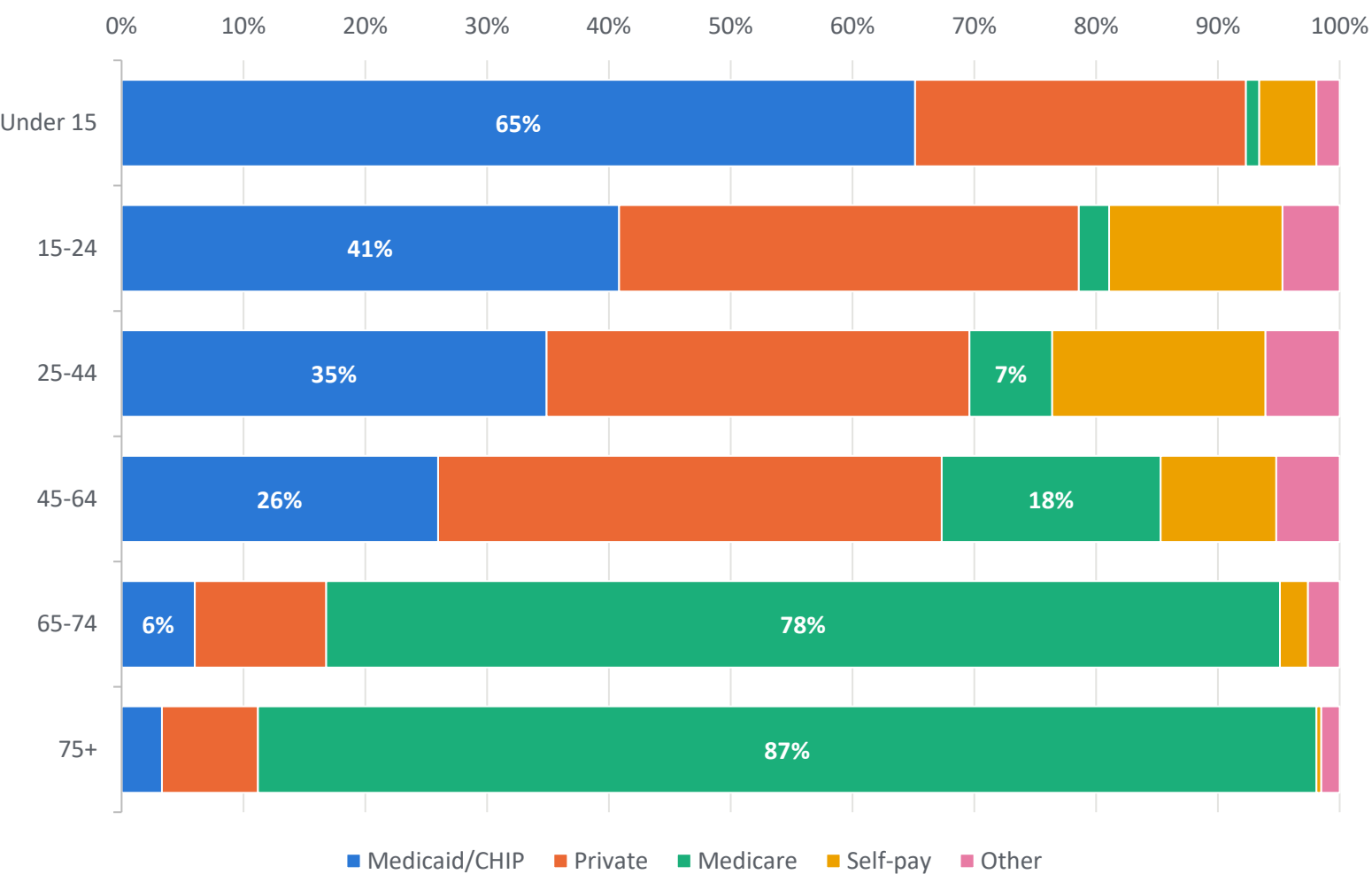
Monday (432k visits/day) vs Sunday (346k), average per day

Staffing

Stagger shifts through the 7-10am ramp, keep full coverage to ~midnight (median stay 2.6 h), and roster heavier on Mondays.

Public payers fund most ED visits, and age decides who pays

Primary expected payer by age group, weighted % of visits with a known payer



54.8%

of known-payer visits paid by Medicaid (34.9%) or Medicare (19.9%)

11.0%

self-pay or charity care; EDs must treat regardless (EMTALA)

Margin pressure

Private insurance (30.9%) cross-subsidizes. 10.8% of visits have no recorded payer, a revenue-cycle gap.

EDs treat complex patients: chronic disease and injuries

47.6%

visits with 1+ chronic condition
(90% at 75+)

33.1%

injury, overdose or adverse-effect visits (7.3% intentional)

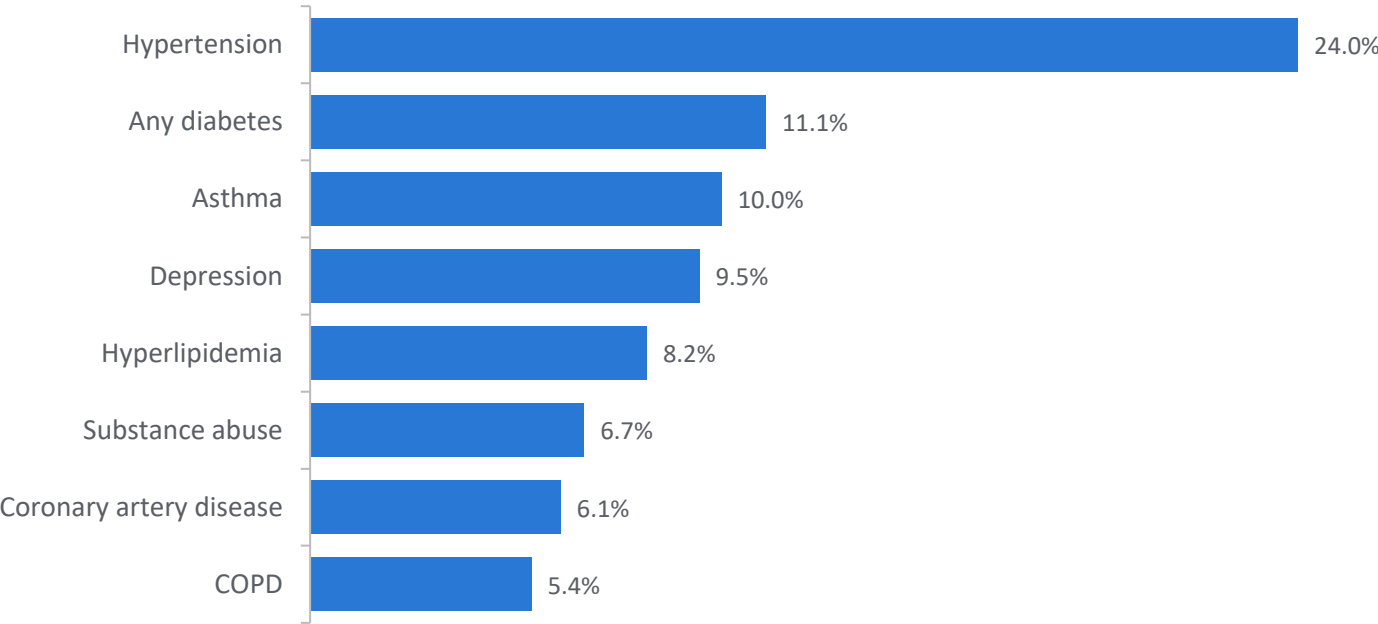
47.0%

visits with imaging; 42.4% with a blood test (CBC in 85%)

37.2%

visits with 3+ medications (mean 2.49, median 2)

Most common chronic conditions (% of ED visits, weighted)



What it means

EDs act as a front line for chronic disease: care coordination and follow-up referrals could move avoidable visits to cheaper settings.

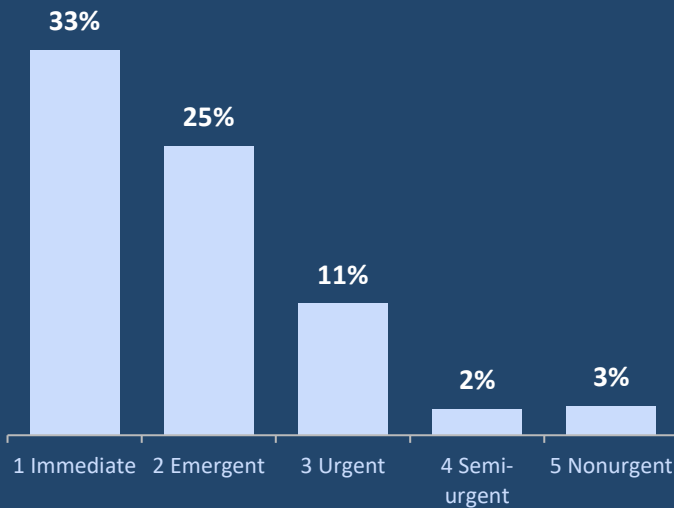
Diabetes is split across three checkboxes; combined it is the #2 condition, not #9.

Intensity follows complexity: imaging rises from 29% of child visits to 70% at 75+, and medications from 2.0 per visit (no chronic condition) to 3.8 (three or more).

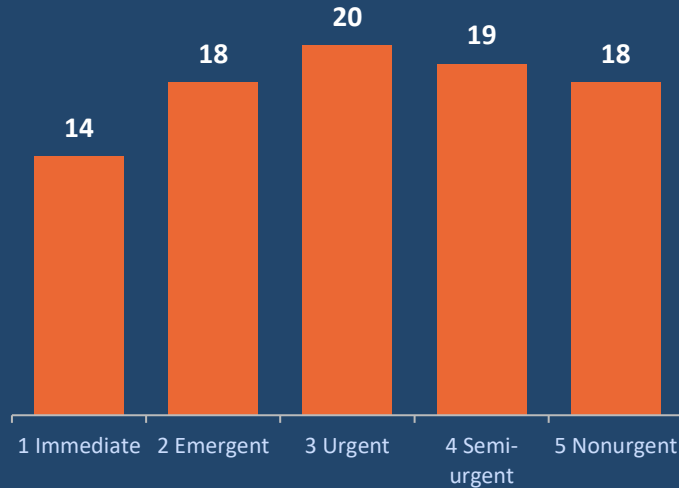
Triage sorts outcomes sharply, but not waiting times

Weighted, by triage level (1 = immediate ... 5 = nonurgent)

Admitted to hospital (%)



Median wait to see a provider (min)



Correlations (weighted r)

blood tests - admitted 0.37

chronic conditions - admitted 0.33

Length of visit - # blood tests 0.31

Age - Medicare 0.61 (eligibility at 65)

Deaths are rare (28 in ED, 7 DOA in sample), so we compare rates: 6.4% of immediate patients die in the ED.

Takeaways for hospital managers

Staff to the plateau and Mondays · Fast-track levels 4-5 so emergent patients stop waiting as long as nonurgent ones · Protect margins through Medicaid contracting and payer capture · Invest in chronic-care coordination